

PURCHASE ORDER ANALYTICS DASHBOARD

SAP ABAP Project Documentation

Project Type: Custom SAP ABAP Report using ALV (CL_SALV_TABLE)

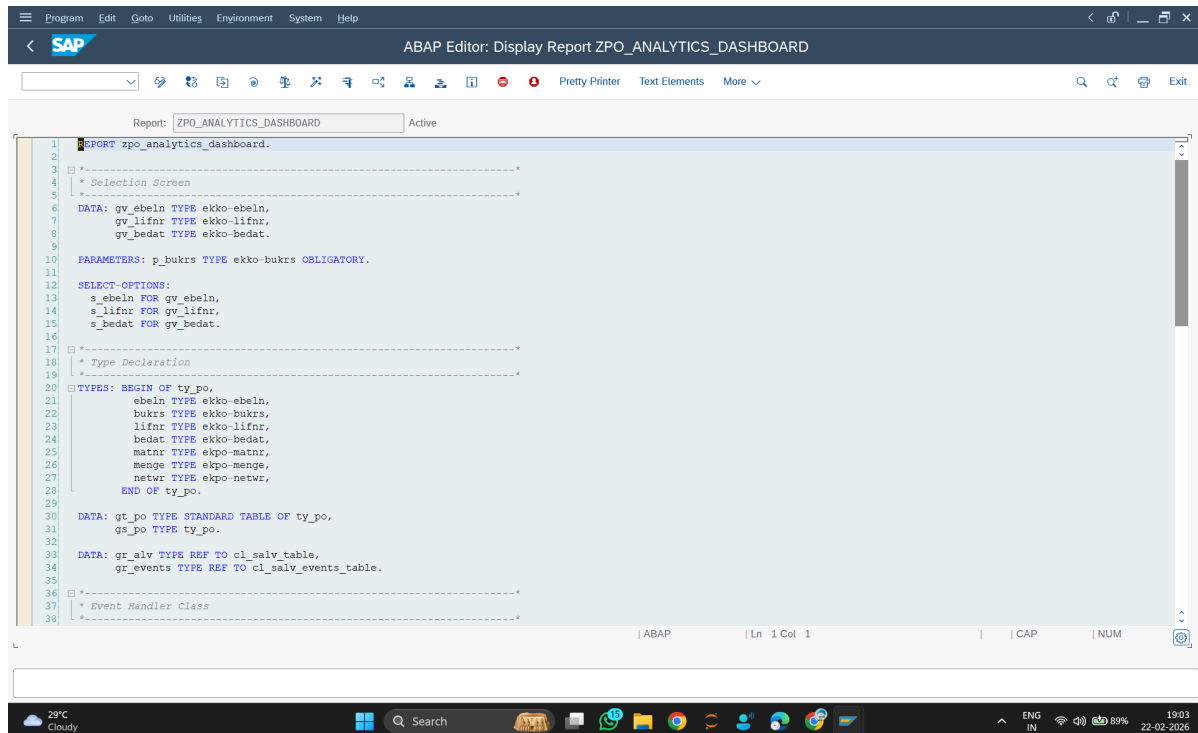
Tables Used: EKKO (Header), EKPO (Item)

Tool: SAP GUI - ABAP Editor

1. Project Development Process

- Step 1: Created custom report ZPO_ANALYTICS_DASHBOARD in ABAP Editor (SE38).
- Step 2: Designed Selection Screen with parameters and select-options.
- Step 3: Defined internal table structure using TYPES.
- Step 4: Retrieved data from EKKO and EKPO tables using SELECT JOIN.
- Step 5: Implemented ALV display using CL_SALV_TABLE class.
- Step 6: Created Variant for execution flexibility.
- Step 7: Executed report and validated output.

ABAP Code Editor View



```
1 REPORT zpo_analytics_dashboard.
2
3 *-----*
4 * Selection Screen
5 *-----*
6 DATA: gv_ebeln TYPE ekko-ebeln,
7       gv_lifnr TYPE ekko-lifnr,
8       gv_bedat TYPE ekko-bedat.
9
10 PARAMETERS: p_bukrs TYPE ekko-bukrs OBLIGATORY.
11
12 SELECT-OPTIONS:
13   s_ebeln FOR gv_ebeln,
14   s_lifnr FOR gv_lifnr,
15   s_bedat FOR gv_bedat.
16
17 *-----*
18 * Type Declaration
19 *-----*
20 TYPES: BEGIN OF ty_po,
21       ebeln TYPE ekko-ebeln,
22       bukrs TYPE ekko-bukrs,
23       lifnr TYPE ekko-lifnr,
24       bedat TYPE ekko-bedat,
25       matr TYPE ekpo-matnr,
26       menge TYPE ekpo-menge,
27       netwr TYPE ekpo-netwr,
28       END OF ty_po.
29
30 DATA: gt_po TYPE STANDARD TABLE OF ty_po,
31       gs_po TYPE ty_po.
32
33 DATA: gr_alv TYPE REF TO cl_salv_table,
34       gr_events TYPE REF TO cl_salv_events_table.
35
36 *-----*
37 * Event Handler Class
38 *-----*
```

Description: This screenshot shows the creation of the custom ABAP report ZPO_ANALYTICS_DASHBOARD. The selection screen parameters, internal table structures, and ALV class references are defined here.

Variant Attributes Configuration

Variant Edit Goto Environment System Help

< SAP

Variant Attributes

Use Screen Assignment

More

Variant Name: P_BUKRS

Description: Execution

Only for Background Processing

Protect Variant

Only Display in Catalog

System Variant (Automatic Transport)

Screen Assignment

	Created	Selection Screen
☐	☒	1000

Fields of selection screen

Technical name

	Dynp	Field na	Ty	Protect fi	Hide field	Hide field *	Do not save valu	Switch GPA	Required fi	Sel. Variable Ty	Opti	Name of Variable (Input Only Using F
☐	1.000	P_BUKRS	P	☐	☐	☐	☐	☐	☒			
☐	1.000	S_EBELN	S	☐	☐	☐	☐	☐	☐			
☐	1.000	S_LIFNR	S	☐	☐	☐	☐	☐	☐			
☐	1.000	S_BEDAT	S	☐	☐	☐	☐	☐	☐			

Description: This screen displays the variant configuration where execution parameters such as company code are saved for repeated usage.

Selection Screen Execution

ProgramEditGotoSystemHelp

Analytics- Dashboard

Save as Variant...

More

Exit

P_BUKRS:0001

S_EBELN:

S_LIFNR:

S_BEDAT:

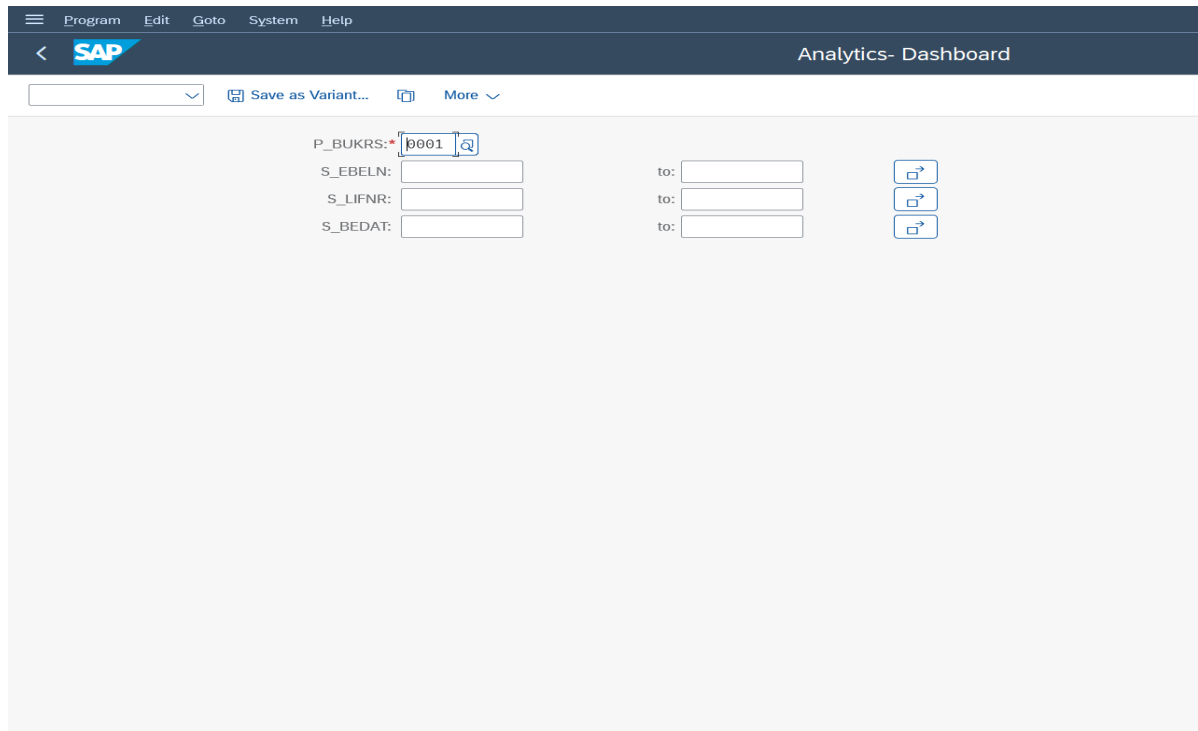
to:

to:

to:

Description: The user enters Company Code and optional filters such as Purchase Document, Vendor, and Date range.

Selection Screen with Company Code 0001



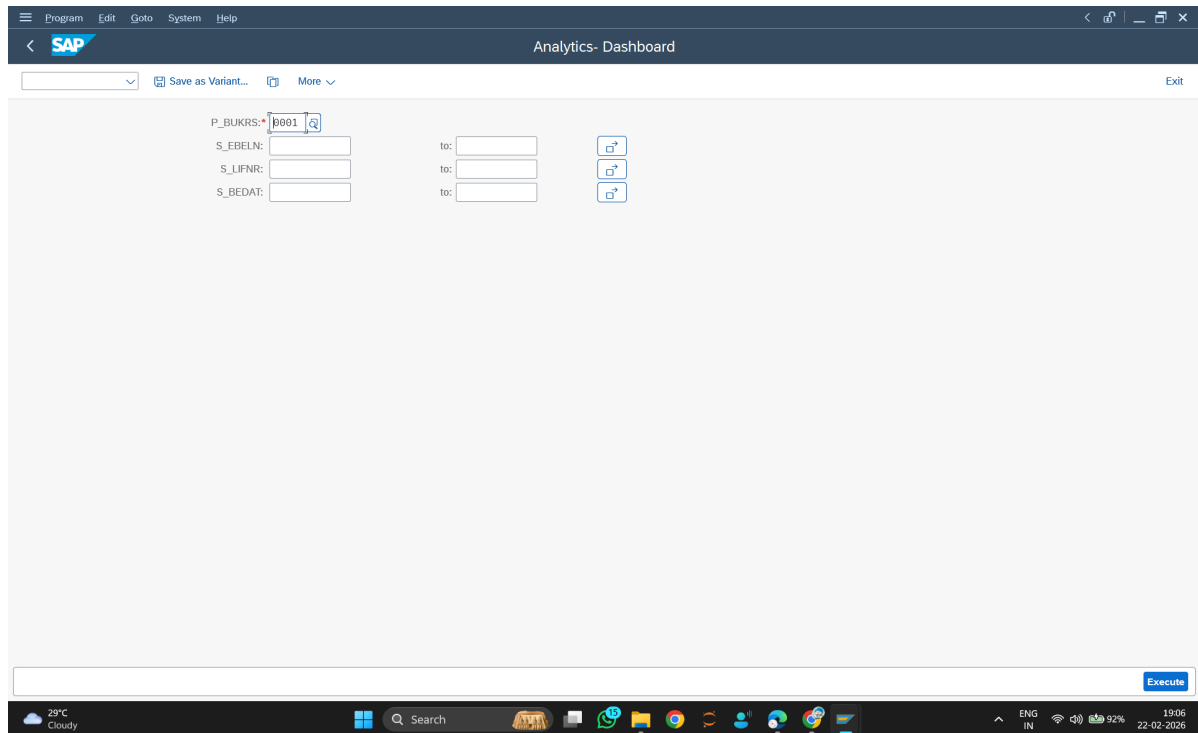
The image shows a screenshot of the SAP Selection Screen for a report. The top bar includes the SAP logo and the text "Analytics- Dashboard". Below the top bar, there is a search bar with a dropdown arrow, and buttons for "Save as Variant...", "More", and a dropdown arrow. The main area contains the following fields:

P_BUKRS:	0001		
S_EBELN:		to:	
S_LIFNR:		to:	
S_BEDAT:		to:	

Each "to:" field has a small icon to its right, likely for date selection or help.

Description: Execution example with Company Code 0001 entered before running the report.

Variant Save Screen



The screenshot displays the SAP 'Variant Save Screen' within the 'Analytics- Dashboard' application. The interface includes a top menu bar with 'Program', 'Edit', 'Goto', 'System', and 'Help' options. Below the menu, there is a toolbar with a dropdown menu, 'Save as Variant...', and 'More' options. The main area contains input fields for 'P_BUKRS' (with a value of '8001'), 'S_EBELN', 'S_LIFNR', and 'S_BEDAT'. To the right of these fields are three 'to:' labels, each followed by a small icon. At the bottom right of the main area is an 'Execute' button. The bottom of the screen shows a Windows taskbar with the date '22-02-2026' and time '19:06'.

Analytics- Dashboard

Save as Variant... More

Exit

P_BUKRS: 8001

S_EBELN:

S_LIFNR:

S_BEDAT:

to:

to:

to:

Execute

29°C Cloudy

Search

ENG IN

92%

19:06

22-02-2026

Description: Variant creation ensures faster execution without re-entering parameters each time.

T001 Table Display

Variant

Edit

Goto

Environment

System

Help

Variant Attributes

Use Screen Assignment

More

Exit

Variant Name: P_BUKRS

Description: Execution

☐ Only for Background Processing

☐ Protect Variant

☐ Only Display in Catalog

☐ System Variant (Automatic Transport)

Screen Assignment

	Created	Selection Screen
☐	☒	1000

Fields of selection screen

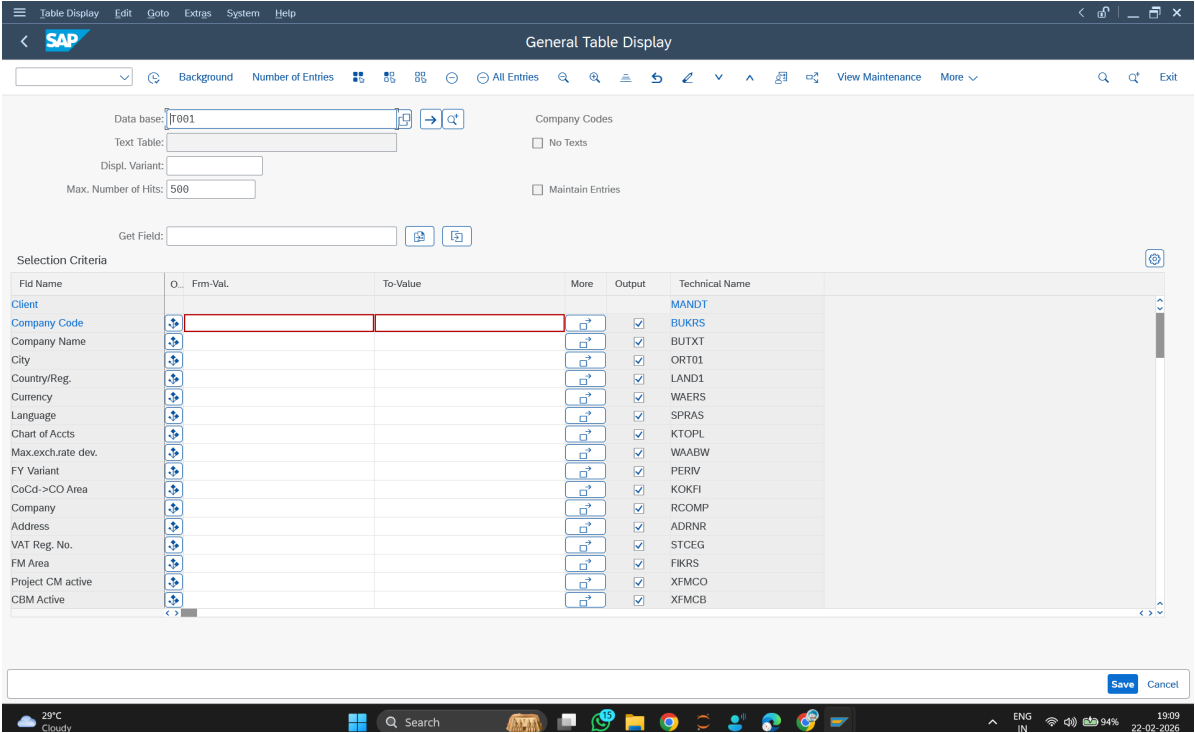
Dyn.	Field no.	Ty.	Protect fi.	Hide fi.	Hide field T.	Do not save valu.	Switch GPA	Required fi.	Sel. Variable Ty.	Opti.	Name of Variable (Input Only Using ...
☐	1.000	P_BUKR	P	☐	☐	☐	☐	☒			
☐	1.000	S_EBELN	S	☐	☐	☐	☐	☐			
☐	1.000	S_LIFNR	S	☐	☐	☐	☐	☐			
☐	1.000	S_BEDAT	S	☐	☐	☐	☐	☐			

Save

Cancel

Description: This screen validates Company Code data from table T001 before running the report.

EKKO Table Entries



Description: EKKO table is used to fetch Purchase Order header details such as document number, company code, vendor.

EKKO Selection Criteria

Table EntryEditGotoSystemHelp

T001: Display of Entries Found

Search in Table: T001Company Codes

Number of Hits: 69

Runtime: 0Maximum No. of Hits: 500

Insert Column:

Details

CoCode	Company Name	City	C-R	Crcy	Langua	Chrt/Ac.	Max ex. d.	FV	C	Co.	Adtre.	VAT Reg. No.	FMA	Project CM	acti.	Acti.	Upd.	Jur. co.	FdsCtrA	Templ.	Trnst P.	Va.	CM.	CM Activa
0001	SAP SE	Walldorf	DE	EUR	DE	INT	10	K4	2															
0003	SAP US (IS-HT-SW)	Palo Alto	US	USD	EN	INT	10	K4	2											X				
0MB1	IS-B Musterbank Deutschl.	Walldorf	DE	EUR	DE	0MB1		K4	2											X				
AE01	Country Template AE	Dubai	AE	AED	EN	INT		K4	2											X				
AR01	Country Template AR	Argentinien	AR	ARS	ES	INT		K4	2											X		2		
AT01	Country Template AT	Austria	AT	EUR	DE	INT	10	K4	2											X		1		
AU01	Country Template AU	Australia	AU	AUD	EN	INT	10	V6	2											X		2		
BE01	Country Template BE	Belgium	BE	EUR	EN	CABE	10	K4	2			BE000009797								X		2		
BR01	Country Template BR	Brazil	BR	BRL	PT	INT	10	K4	2											X		2		
CA01	Country Template CA	Canada	CA	CAD	EN	CANA	10	K4	2											X		2		
CH01	Country Template CH	Switzerland	CH	CHF	DE	CACH	10	K4	2											X		1		
CL01	Country Template CL	Chile	CL	CLP	ES	INT	10	K4	2											X		2		
CN01	Country Template CN	China	CN	CNY	ZH	CACN	10	K1	2											X		2		
CO01	Country Template CO	Colombia	CO	COP	ES	CACO	10	K4	2											X		2		
CZ01	Country Template CZ	Czech Republic	CZ	CZK	CS	CACZ	5	K4	2			CZ1234567890								X				
DE01	Country Template DE	Germany	DE	EUR	DE	GKR	10	K4	2											X				
DE02	Country Template DE	Germany	DE	EUR	DE	IKR	10	K4	2											X				
DK01	Country Template DK	Denmark	DK	DKK	DA	INT	10	K4	2											X				

29°C Cloudy

Search

ENG IN

19:09 22-02-2026

Description: Selection criteria for Purchase Document and Company Code used during development testing.

Selection Screen Final Execution

Pur. Doc.	CoCode	C	Type	Ctl	E S	Created On	Created By	Last Changed It.	Lit.	Suppli.	Langua.	Pyt Ter.	Payment.	Payment.	Payment.	CD ..	CD ..	PORG	PGR	Grcy	Exch. R.	Fix Doc.	Date	VP St.
☐ 4500000000	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.103.412,7443460	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000001	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.102.253,5112510	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000002	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.103.155,0427130	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000003	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.103.401,0229670	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000004	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.102.906,1894720	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000005	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.102.927,4470450	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	
☐ 4500000006	NSPL	F	NB		9	29.01.2026	YASWANTH	20.260.129.103.002,8372440	10	10	1001	EN		0	0	0	0,000	0,000	NSPO	NPG	INR	1,00000	29.01.2026	

Description: Final execution screen before displaying ALV output.

ALV Output Display

Table DisplayEditGotoExtrasSystemHelp

General Table Display

BackgroundNumber of EntriesAll Entries

SearchFilterSort

Exit

Data base: EKKO

Text Table:

Displ. Variant:

Max. Number of Hits: 500

Get Field:

Purchasing Document Header

☐ No Texts

☐ Maintain Entries

Selection Criteria

Fld Name	O.	Frm-Val.	To-Value	More	Output	Technical Name
Client						MANDT
Purchasing Doc.					☒	EBELN
Company Code					☒	BUKRS
Doc. Category					☒	BSTYP
Document Type					☒	BSART
Control					☒	BSAKZ
Del. Indicator					☒	LOEKZ
Status					☒	STATU
Created On					☒	AEDAT
Created By					☒	ERNAM
Last Changed					☒	LASTCHANGEDATETIME
Item Interval					☒	PINCR
Last Item					☒	LPONR
Supplier					☒	LIFNR
Language					☒	SPRAS
Pyt Terms					☒	ZTERM
Payment in					☒	ZBD1T

Save

Cancel

Description: ALV Grid output displaying Purchase Document, Supplier, Material, Order Quantity, and Net Order Value.

ALV Output (Extended View)

ProgramEditGotoSystemHelp

Analytics- Dashboard

Save as Variant...

More

Exit

P_BUKRS:*

NSPL

S_EBELN:

S_LIFNR:

S_BEDAT:

to:

to:

to:

Execute

Description: Extended ALV view showing multiple purchase orders with proper formatting.

Final Output Validation

ListEditGotoSettingsSystemHelp

Analytics- Dashboard

More

Exit

Purchasing Doc.	CoCo	Supplier	Document Date	Material	Order Quantity	Net Order Value
4500000001	NSPL	1001	29.01.2026	8	10,000	100,00
4500000004	NSPL	1001	29.01.2026	12	100,000	10,000,00
4500000005	NSPL	1001	29.01.2026	13	100,000	10,000,00
4500000006	NSPL	1001	29.01.2026	8	1,000,000	10,000,00
4500000002	NSPL	1001	29.01.2026	9	100,000	10,000,00
4500000003	NSPL	1001	29.01.2026	10	100,000	10,000,00
4500000000	NSPL	1001	29.01.2026	8	110,000	1,100,00

Description: Final validated output confirming successful integration between EKKO and EKPO tables.

Conclusion

The Purchase Order Analytics Dashboard was successfully developed using SAP ABAP. The project demonstrates strong understanding of selection screens, internal tables, database joins, and ALV reporting techniques using object-oriented ABAP.